

Advanced Optimization

Chapter 17 — Penalty and Augmented Lagrangian Methods

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Fall 2026

Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 17

One question, three answers

The question: can we trade the constrained problem $\min f$ s.t. $c_i = 0$ ($i \in \mathcal{E}$), $c_i \geq 0$ ($i \in \mathcal{I}$) for a sequence of *unconstrained* (or bound-constrained) problems?

	Smooth?	Exact (finite μ)?	Price paid
Quadratic penalty (§17.1)	yes	no : needs $\mu \rightarrow \infty$	ill-conditioning
ℓ_1 penalty (§17.2)	no	yes: $\mu > \ \lambda^*\ _\infty$	nonsmooth subproblems
Augmented Lagrangian (§17.3–4)	yes	effectively: $\mu \geq \bar{\mu}$ finite	must track λ

Why this chapter sits here

Ch. 11 already used a manufactured merit function — and met its traps. Here the idea becomes a *method*: penalties are the merit functions that will globalize SQP (Ch. 18), the quadratic penalty is the regularizer hiding inside SQP's linear algebra, and the log barrier of Ch. 19 is this chapter's smooth cousin for inequalities. The augmented Lagrangian is also the seed of ADMM — one slide at the end.

Running example throughout: the Ch. 12 circle, $\min x_1 + x_2$ s.t. $x_1^2 + x_2^2 = 2$, with $x^* = (-1, -1)^T$, $\lambda^* = -\frac{1}{2}$.

Quadratic penalty: square the violations and crank μ

$$Q(x; \mu) = f(x) + \frac{\mu}{2} \sum_{i \in \mathcal{E}} c_i^2(x) \quad \left(+ \frac{\mu}{2} \sum_{i \in \mathcal{I}} ([c_i(x)]^-)^2 \text{ for inequalities} \right). \quad (17.2, 17.7)$$

Framework 17.1: for $\mu_k \uparrow \infty$, minimize $Q(\cdot; \mu_k)$ to tolerance $\tau_k \downarrow 0$, warm-starting from x_{k-1} ; adapt μ_{k+1} .

Example 17.1 (circle): $\mu = 1$ gives minimizer $\approx (-1.1, -1.1)$, infeasible; $\mu = 10$: a sharp trough hugs the circle — the narrowing trough *is* the ill-conditioning arriving. Warnings: Q can be **unbounded below** for nice problems ($\min -5x_1^2 + x_2^2$ s.t. $x_1 = 1$: unbounded for all $\mu < 10$); the inequality term $([c]^-)^2$ makes Q only C^1 .

Convergence (Theorems 17.1–17.2)

17.1: exact *global* minimizers of each subproblem $\Rightarrow$ every limit point is a global solution. (Five lines: $Q(x_k; \mu_k) \leq Q(\bar{x}; \mu_k) = f(\bar{x})$ bounds $\sum c_i^2(x_k) \leq \frac{2}{\mu_k} [f(\bar{x}) - f(x_k)] \rightarrow 0$.)

17.2: inexact minimizers ($\tau_k \rightarrow 0$): limit points are either **stationary points of $\|c\|^2$** (the Ch. 11 trap, verbatim) or, under independent constraint gradients, KKT points — with a gift:

$$-\mu_k c_i(x_k) \rightarrow \lambda_i^*. \quad (17.10)$$

The penalty method estimates multipliers *for free*. Hold that thought for §17.3.

III-conditioning, and the reformulation that changes the story

Near the subproblem minimizer,

$$\nabla_{xx}^2 Q(x; \mu_k) \approx \nabla_{xx}^2 \mathcal{L}(x, \lambda^*) + \mu_k A(x)^T A(x) \quad (17.19)$$

— a μ -independent part plus a rank- $|\mathcal{E}|$ part with eigenvalues $O(\mu_k)$: condition number $O(\mu_k) \rightarrow \infty$. Newton systems (17.20) become numerically hopeless precisely when the method needs them.

The fix (17.21): un-eliminate the multiplier

Set $\zeta = \mu A(x)p$; the Newton step also solves

$$\begin{bmatrix} \nabla^2 f(x) + \sum_i \mu_k c_i(x) \nabla^2 c_i(x) & A(x)^T \\ A(x) & -(1/\mu_k)I \end{bmatrix} \begin{bmatrix} p \\ \zeta \end{bmatrix} = \begin{bmatrix} -\nabla_x Q \\ 0 \end{bmatrix}$$

— bounded entries, KKT shape (Ch. 16!), condition number tamed.

- Read it twice: it is the quadratic penalty's Newton step, *and* it is a **regularized SQP step** — the $-(1/\mu)I$ block keeps the system nonsingular even for rank-deficient A . One method, two theories; Ch. 18 will meet it from the other side.
- The residual moral: for small μ the step ignores the constraint linearization (slow progress to feasibility); for large μ conditioning bites. The quadratic penalty has **one dial and two failure modes** — the setup for §17.3.

The ℓ_1 penalty: nonsmooth, and exact

$$\phi_1(x; \mu) = f(x) + \mu \sum_{i \in \mathcal{E}} |c_i(x)| + \mu \sum_{i \in \mathcal{I}} [c_i(x)]^-. \quad (17.22)$$

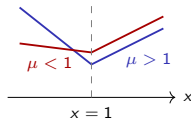
Theorem 17.3 (exactness)

If x^* is a strict local solution with multipliers λ^* , then for every $\mu > \|\lambda^*\|_\infty$, x^* is a local minimizer of $\phi_1(\cdot; \mu)$. One minimization, finite μ , exact answer — no $\mu \rightarrow \infty$, no systematic infeasibility.

Theorem 17.4 (converse)

Stationary points of ϕ_1 (in the directional-derivative sense) for all $\mu > \hat{\mu}$: if feasible $\Rightarrow$ KKT point (via *Farkas* — Ch. 12's lemma, third appearance); if infeasible $\Rightarrow$ stationary for the violation measure h . Same honest caveat as always.

On the circle problem: any $\mu > |\lambda^*| = 0.5$ makes the ϕ_1 -minimizer exactly x^* — contours show sharp corners along the feasible circle (Figure 17.4): the kinks are where the exactness lives.



Example 17.2: $\min x$ s.t. $x \geq 1$ ($\lambda^* = 1$). ϕ_1 has its kinked minimizer at $x^* = 1$ iff $\mu > 1$; below the threshold it just slides away.

Minimizing ϕ_1 in practice: linearize inside, and it becomes a QP

Framework 17.2 (classical): minimize $\phi_1(\cdot; \mu_k)$, raise μ ($\times 5$ or 10) until the violation $h(x_k) \leq \tau$. The dial dilemma remains (μ_0 too small: many wasted cycles; too large: hard subproblems) — but now a *finite* correct μ exists.

The working model (17.30): don't fight the kinks, structure them

$$q(p; \mu) = f + \nabla f^T p + \frac{1}{2} p^T W p + \mu \sum_{\mathcal{E}} |c_i + \nabla c_i^T p| + \mu \sum_{\mathcal{I}} [c_i + \nabla c_i^T p]^-$$

Elastic variables $r, s, t \geq 0$ absorb the absolute values $\Rightarrow$ a **smooth QP** (Ch. 16 solves it). Minimizing ϕ_1 via these models *is* an SQP method with built-in feasibility relaxation — Ch. 18 calls it Sl_1QP (Fletcher).

- ▶ In the wild: SNOPT uses the ℓ_1 “elastic mode” as its safeguard when the QP subproblem goes infeasible/unbounded; KNITRO ships an ℓ_1 method with LP subproblems.
- ▶ The ℓ_1 penalty extends the reach of everything else: MPCCs (complementarity constraints violate every standard CQ) become tractable by *penalizing* the offending constraints instead of linearizing them.
- ▶ General norms: exactness holds with $\mu > \|\lambda^*\|_D$ (dual norm) — $\ell_1, \ell_\infty, \ell_2$ (unsquared!) all work; **squaring is exactly what destroys exactness**.

Augmented Lagrangian: fix the penalty's bias instead of overpowering it

Theorem 17.2 says the quadratic-penalty minimizer is *predictably* wrong: $c_i(x_k) \approx -\lambda_i^*/\mu_k$ (17.35). If the error is systematic, subtract it:

$$\mathcal{L}_A(x, \lambda; \mu) = f(x) - \sum_{i \in \mathcal{E}} \lambda_i c_i(x) + \frac{\mu}{2} \sum_{i \in \mathcal{E}} c_i^2(x) \quad (17.36)$$

— Lagrangian + penalty, hence the name. Stationarity of $\mathcal{L}_A(\cdot, \lambda^k; \mu_k)$ now gives

$$\lambda_i^* \approx \lambda_i^k - \mu_k c_i(x_k) \implies c_i(x_k) \approx -\frac{\lambda_i^* - \lambda_i^k}{\mu_k}. \quad (17.38)$$

Read (17.38) slowly — it is the whole section

Infeasibility $\propto \|\lambda^k - \lambda^*\|/\mu_k$, not $\|\lambda^*\|/\mu_k$: **two dials now** — improve the multiplier estimate *or* raise μ . And (17.38) hands us the multiplier update for free:

$$\lambda^{k+1} = \lambda^k - \mu_k c(x_k) \quad (17.39)$$

Framework 17.3 = Framework 17.1 + this one line. (First-order multiplier update; “method of multipliers”, Hestenes 1969 / Powell 1969.)

Example 17.4 (circle, same $\mu = 1$ as before): with $\lambda^k = -0.4$ (a mediocre estimate of $\lambda^* = -0.5$), the minimizer of $\mathcal{L}_A$ is $\approx (-1.02, -1.02)$ — versus the penalty's $(-1.1, -1.1)$. *Same conditioning, five times less infeasibility.* The multiplier term is doing work μ never has to do.

Why it works at finite μ (Theorems 17.5–17.6)

Theorem 17.5 (exact multiplier $\Rightarrow$ exact minimizer, finite μ)

LICQ + second-order sufficiency at $(x^*, \lambda^*) \Rightarrow$ there is $\bar{\mu}$ such that for all $\mu \geq \bar{\mu}$, x^* is a strict local minimizer of $\mathcal{L}_A(\cdot, \lambda^*; \mu)$.

Skeleton: the gradient vanishes for every μ ; if no $\bar{\mu}$ worked, unit w_k with $w_k^T(\nabla^2 \mathcal{L} + kA^T A)w_k \leq 0$ would force $\|Aw_k\| \rightarrow 0$, and a limit w ($Aw = 0$, $w^T \nabla^2 \mathcal{L} w \leq 0$) contradicts Thm 12.6 on the critical cone.

Theorem 17.6 (Bertsekas): the realistic version, with rates

For $\|\lambda^k - \lambda^*\| \leq \mu_k \delta$ and $\mu_k \geq \bar{\mu}$: the subproblem has a unique local solution x_k near x^* with

$$\|x_k - x^*\| \leq M \|\lambda^k - \lambda^*\| / \mu_k, \quad \|\lambda^{k+1} - \lambda^*\| \leq M \|\lambda^k - \lambda^*\| / \mu_k, \quad (17.45\text{--}17.46)$$

and the subproblem Hessian is positive definite (standard unconstrained machinery applies).

(17.46): the λ -iteration converges linearly with ratio M/μ_k — any fixed $\mu > M$ suffices; larger μ buys speed, not correctness. **No ill-conditioning is required for convergence** — the sentence the quadratic penalty could never say. Practice keeps μ moderate, raising it only when feasibility stalls.

LANCELOT (Algorithm 17.4): the bound-constrained Lagrangian in production

Inequalities via slacks: $c_i(x) - s_i = 0, s_i \geq 0$ (17.47) $\Rightarrow$ everything is

$$\min f(x) \quad \text{s.t.} \quad c(x) = 0, \quad l \leq x \leq u. \quad (17.48)$$

Augment *only the equalities*; keep the bounds explicit in the subproblem: $\min_{l \leq x \leq u} \mathcal{L}_A(x, \lambda; \mu)$ (17.50), stationarity measured by the projection residual $\|x - P(x - \nabla_x \mathcal{L}_A, l, u)\|$ (17.51–17.52).

The two-branch outer loop (Algorithm 17.4)

Solve the subproblem to tolerance ω_k ; then test feasibility $\|c(x_k)\| \leq \eta_k$:

- ▶ **good enough** $\rightarrow$ update $\lambda^{k+1} = \lambda^k - \mu_k c(x_k)$, keep μ , *tighten* η, ω ;
- ▶ **not good enough** $\rightarrow \mu \times 100$, **do not touch λ** (feasibility first), reset tolerances.

Two dials, two branches — (17.38) turned into control logic.

- ▶ Inner solver: gradient projection + trust region on a quadratic model (17.54) — **Ch. 16's Algorithm 16.5**: Cauchy point on the bent path, then CG on the free variables. No KKT or Jacobian factorizations, **matrix–vector products only** — why LANCELOT scales.
- ▶ Hessian can be BFGS/SR1, partially separable structure exploited (Ch. 6–7 collect again; SR1's promised natural habitat is exactly this trust-region subproblem).

Cousins and verdicts (§17.4–17.5)

- ▶ **LCL / MINOS:** minimize the (augmented) Lagrangian subject to *linearized* constraints — keeps the subproblem linearly constrained, superlinear near the solution. With LANCELOT, one of the two flagship AL implementations.
- ▶ **Unconstrained formulation for inequalities:** the inner $\max_{\lambda \geq 0}$ with a proximity penalty can be done in closed form (17.63), leaving smooth piecewise terms $\psi(c_i, \lambda_i; \mu)$ (17.64–17.65) — elegant; adopted by no major solver.
- ▶ **Where each idea lives now:** quadratic penalty $\rightarrow$ regularization inside SQP linear algebra (17.21), rarely standalone; $\ell_1 \rightarrow$ elastic/safeguard modes of SNOPT and KNITRO, and the MPCC workhorse; augmented Lagrangian $\rightarrow$ LANCELOT, MINOS — and, by splitting, ADMM (next slide).
- ▶ Choosing in 2026: AL shines when subproblem solvers are strong and factorizations are out of reach; SQP/interior-point (Ch. 18–19) dominate when they are not.

Beyond the book: from Framework 17.3 to ADMM

Not in N&W — a 2026 postscript the chapter deserves

Take an objective that *splits*, couple the pieces by a linear constraint:

$$\min_{x,z} f(x) + g(z) \quad \text{s.t. } Ax + Bz = c, \quad \mathcal{L}_A = f(x) + g(z) + \lambda^T (Ax + Bz - c) + \frac{\mu}{2} \|Ax + Bz - c\|^2.$$

ADMM = Framework 17.3 with the joint x, z -minimization replaced by *one sweep* of block coordinate minimization:

$$x^{k+1} = \arg \min_x \mathcal{L}_A(x, z^k, \lambda^k); \quad z^{k+1} = \arg \min_z \mathcal{L}_A(x^{k+1}, z, \lambda^k); \quad \lambda^{k+1} = \lambda^k + \mu(Ax^{k+1} + Bz^{k+1} - c).$$

Same multiplier update (17.39), same two-dial logic — the alternation is the entire novelty.

- ▶ Why it took over: splitting makes each block *easy* (closed-form prox steps, projections) and parallel — consensus form distributes across machines or regions. Boyd et al. (2011); convex case converges for any fixed $\mu > 0$.
- ▶ Power-systems incarnation: *distributed OPF* — regions solve local OPFs while λ prices boundary-flow consistency; the multiplier is once more an LMP-like price.
- ▶ Caveats: slow tail (high accuracy is not its game); nonconvex guarantees thin. Separate handout planned for the ADMM week.

Chapter 17 in one table

	Quadratic penalty	ℓ_1 penalty	Augmented Lagrangian
Function	$f + \frac{\mu}{2} \ c\ ^2$	$f + \mu \ c\ _1$	$f - \lambda^T c + \frac{\mu}{2} \ c\ ^2$
Smooth	yes (C^1 w/ ineq.)	no (kinks at $c_i = 0$)	yes
Exact	no: $c \approx -\lambda^*/\mu$	yes: $\mu > \ \lambda^*\ _\infty$	yes at finite $\bar{\mu}$, given λ^*
Needs $\mu \rightarrow \infty$	yes	no	no; rate M/μ in λ
Conditioning	$O(\mu)$, fatal	n/a (nonsmooth)	moderate μ suffices
Infeasible trap	stat. pts of $\ c\ ^2$	infeasible stat. pts of h	same family
Modern role	regularizes SQP (17.21)	$S\ell_1$ QP, elastic modes, MPCC	LANCELOT/MINOS; ADMM seed

Three ideas to carry forward

The quadratic penalty's error is systematic, $c \approx -\lambda^/\mu$ — and a systematic error can be subtracted: that subtraction is the augmented Lagrangian. Exactness needs nonsmoothness (unsquared norms) or multipliers; squaring buys smoothness and pays with $\mu \rightarrow \infty$.*

$\lambda^{k+1} = \lambda^k - \mu c(x_k)$: one line converts a penalty method into a multiplier method — and, with splitting, into ADMM.

Homework (1/2)

Derivations

- 1 Verify that $Q(x; \mu)$ for problem (17.5) is unbounded below exactly when $\mu < 10$, and bounded (with minimizer $\rightarrow (1, 0)$) when $\mu > 10$.
- 2 Fill in the proof of Theorem 17.1, and explain in one sentence why the argument needs *global* subproblem minimizers.
- 3 Derive the equivalence of (17.20) and the augmented system (17.21) via $\zeta = \mu A(x)p$, and identify the block that regularizes rank-deficient A .
- 4 For Example 17.2, compute $D(\phi_1(1; \mu); p)$ for $p \geq 0$ and $p < 0$, and locate the exactness threshold. Then verify $\mu > 0.5$ works for the circle problem.
- 5 Derive the multiplier update (17.39) from the stationarity comparison (17.37) $\leftrightarrow$ (17.17), and show how (17.38) yields the two error bounds of Theorem 17.6 heuristically.
- 6 Carry out the inner maximization (17.62) $\rightarrow$ (17.63) $\rightarrow$ (17.65) explicitly for one inequality constraint.

Homework (2/2)

Implementation

7. *The bias experiment.* On the circle problem, run Framework 17.1 with fixed $\mu = 10$ and Framework 17.3 with fixed $\mu = 10$ (update λ by (17.39)). Plot $|c(x_k)|$ vs. k : penalty stalls at $|\lambda^*|/\mu = 0.05$; AL drives it to zero. One plot, the whole chapter.
8. *The conditioning experiment.* Compute $\kappa(\nabla_{xx}^2 Q)$ at the subproblem minimizers for $\mu = 10^1, \dots, 10^8$; confirm $O(\mu)$ growth, then repeat for $\mathcal{L}_A$ with the (17.39)-updated λ at fixed $\mu = 10$.
9. Implement a miniature Algorithm 17.4 (outer loop only; inner solve via `scipy` L-BFGS-B) for a small problem with bounds; log which branch fires each iteration.
10. *Optional (ADMM):* solve consensus least squares $\min \sum_j \|A_j x_j - b_j\|^2$ s.t. $x_j = z$ by ADMM; watch primal/dual residuals and the effect of μ .

Next: Ch. 18 — *sequential quadratic programming*.